

Beyond the Heatmap: Toward Metaphorical Visualization of Embodied Carbon in Architectural Design

Prompting History

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This document details the AI-driven prototyping process for the metaphorical visualization (MET-VIS) proposal for embodied carbon (EC), as developed in the article 'Beyond the Heatmap: Toward Metaphorical Visualization of Embodied Carbon in Architectural Design.' This process constitutes Phase 2 of the research methodology.

Functional Requirements

The selection of AI as a prototyping tool enabled the simultaneous processing of text and images, prioritizing creative interpretation over photorealistic fidelity. Ideally, a reference image setup and in-process hand-drawing tools were considered to preserve human agency for future study stages.

AI Tools Employed in the Process

Two artificial intelligence models were employed as co-creative partners throughout the prototyping process: Gemini and Adobe Firefly. Gemini served as a support tool for structuring prompts and optimizing character counts when necessary. Adobe Firefly was utilized for image generation, employing the 'Gemini 3.1 w / Nano Banana 2' model configuration.

1. Preliminary Exploratory Test

To explore the workflow and evaluate the generation engine, an initial test was conducted using hand-drawn sketches.

Initially, we hand-drew a basic colored 3D volume to showcase and highlight different interior architectural components: walls, floor, ceiling, window, and doors, and to have immediate and full control over the image.

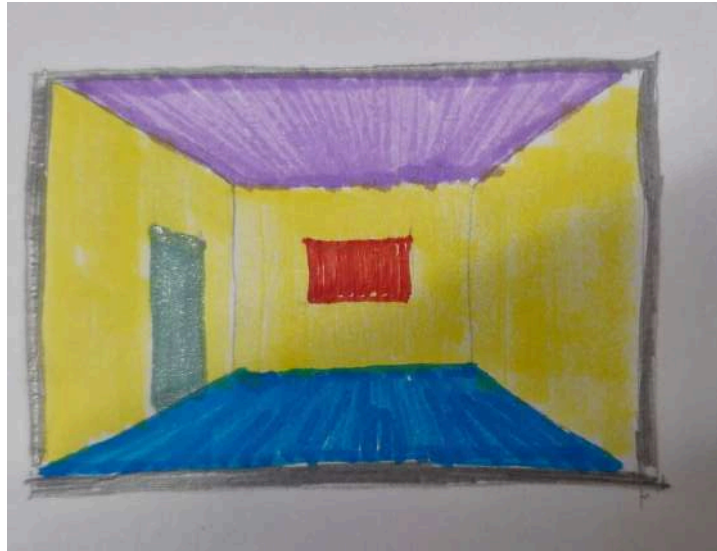


Figure 1. Hand-sketched 3D interior perspective. Source: Authors (2026)

Then, we sketched our initial proposal for MET-VIS and advanced by uploaded the two drawings to Gemini along with a contextualized description to set a starting point for AI's first version of the prompt.



Figure 2. Hand-sketched preliminary proposal for MET-VIS. Source: Authors (2026)

Initial Conceptualization

Create a text-based prompt to describe metaphorical visualizations (MET-VIS), which is a type of visualization more capable of promoting greater cognitive fluency by evoking multisensory associations to support evidence-based design.

The <ULTIMATE GOAL> is to speculate on alternative visual strategies that make objective data more tangible, particularly for non-experts.

Below, I will present the CONTEXT (embodied carbon of architectural components), INSTRUCTION A (about figure 1 attached), and INSTRUCTION B (about figure 2 attached) for you to follow in sequence and create a text-based prompt to describe metaphorical visualizations (MET-VIS).

Then I will add the graphical structure directions: METAPHOR (for your global visual approach) and VISUAL LANGUAGE (basic elements and combination forms you have to combine).

This text-based prompt has to indicate <empty spaces>, the <placeholders> to act as filling patterns to indicate BASIC ELEMENTS to be used, and COMBINATION FORMS.

CONTEXT

<Embodied Carbon> is a <massive> environmental challenge characterized by <high-intensity>, carbon-intensive <impacts> across a <building's> lifecycle. The concept <requires balancing> a strict ecological carbon budget by <tracking two distinct phenomena> as STATIC MATERIAL PERFORMANCE METRICS (mass-scale emissions/breakdowns) and DYNAMIC MITIGATION/SEQUESTRATION ACTION (radical reduction pathways).

INSTRUCTION A

The <1st image attached> shows a 3D hand-sketch model of the interior of a room. We can see:

WALLS: yellow

CEILING: purple

FLOOR: blue

DOOR: green

WINDOW: red

Each color has to be respected for each architectural component.

This room represents the basic status of the room: without EC presentation.

INSTRUCTION B

The <2nd image attached> shows the transformation of the 3D model of the interior of a room. We can see:

<deformed> WALLS: yellow - The amoebic walls turned into a blob, where the weight of the material is concentrated in the bottom part, as in a gravity reaction.

<deformed> CEILING: purple - Pending amoebic format.

<deformed> FLOOR: blue - Hole in the center of the room.

<deformed> DOOR: green - Spikes

<deformed> WINDOW: red - Spikes

For the deformations: the higher the EC value is, the higher the deformation becomes.

METAPHOR

To superimpose two representational layers, the best approach is to use a Composite metaphor;
Carbon budgets (abstract information): Biomorphic Metaphor to represent a morphing shape/parasitic behavior);

Growth & Spread: Carbon data borrows natural, fluid forms (resembling an organism or parasite) that physically engulf or expand past architectural boundaries based on environmental severity.

Weight and Hierarchy: Employs environmental attributes (mass, gradient, density) to generate immediate cognitive effects regarding data importance and severity

Spatial architectural (tangible features): Metaphorical Worlds to indicate the room and components as containers;

3D Structure: a fully dimensional spatial environment where the visual representations are in 3D;

Component-Boundary Alignment: Every architectural component maintains a unique representation where its graphic representation boundaries explicitly conform to the physical boundaries of the structural element itself;

Holistic Superimposition: Layered, distinct data-driven forms, enabling holistic perception (all quantitative metrics and physical parts are recognized simultaneously);

VISUAL LANGUAGE

Visual language (drawing principles):

Basic Elements:

Volume and Form: Volumetric scaling handles tangible metrics in structural metaphors (e.g., larger volumes signify higher material-specific carbon intensity).

Plane and Position: Explicitly matches physical architectural boundaries, anchoring data to components like walls, slabs, or door and window frames.

Color, Texture, and Direction: Individually highlight data tiers and intensity vectors.

Combination Forms:

Grouping and Balance: Adjusts visual hierarchy, creating a heavy, distorted visual "burden" for high-impact phases or light, stable representations for sequestered carbon.

Transformation (Subtraction/Addition): Employs progressive expansion (amoebic addition) to track physical accumulation upfront, or receding/subtractive forms to show mitigation strategies.

Following this, the initial conceptualization was refined and structured with the assistance of AI processing, resulting in the prompt presented below.

AI-refined prompt

[**OBJECTIVE**]: Generate an <IMAGE> USING THE ATTACHED IMAGE AS BASELINE LAYER, for 3D metaphorical data visualization (MET-VIS) mapping abstract **Embodied Carbon (EC)** metrics onto a physical architectural container to bridge the affective gap and promote high cognitive fluency for non-experts.

1. Spatial Architectural Layer (Baseline Reality Container)

- **Spatial Environment:** A fully dimensional, 3D interior spatial environment representing a single room.
- **Component Boundary & Color Alignment:** Maintain strict **<Plane and Position>** and **<Color and Texture>** component-boundary alignment where each element adheres to its original color scheme and structural position:
 - **WALLS:** Painted **yellow**, establishing vertical structural boundaries.
 - **CEILING:** Colored **purple**, establishing the upper overhead enclosure.
 - **FLOOR:** Colored **blue**, establishing the horizontal base support plane.
 - **DOOR:** Colored **green**, framed as a distinct rectangular threshold.
 - **WINDOW:** Colored **red**, framed as an opening structural aperture.

2. Biomorphic Data Overlay (The Parasitic Metaphor)

Superimpose a dynamic, data-driven biomorphic metaphor over the structural baseline to represent the **<Massive environmental impact of Embodied Carbon>**. The intensity of visual deformation scales directly with numerical environmental severity (**Higher EC Value = Higher Metaphorical Deformation**).

Statical Material Performance Metrics (Mass-Scale Accumulation)

Apply **[COMBINATION FORM: Transformation via Addition/Expansion]** to represent physical carbon accumulation using parasitic, amoebic forms:

- **YELLOW WALLS:** Morph into an active **<Volume and Form>** blob. Apply **[COMBINATION FORM: Grouping and Balance]** to concentrate the visual mass and density at the bottom part, simulating a heavy gravity reaction.
- **PURPLE CEILING:** Deform into a pending, hanging amoebic format that drops downward based on **<Direction>** **[COMBINATION FORM: Grouping and Balance]** vectors of environmental burden.
- **GREEN DOOR:** Structural boundary breaks outward into sharp, aggressive biomorphic **<Texture>** spikes.
- **RED WINDOW:** Boundary deforms into a matching array of sharp biomorphic **<Texture>** spikes.

Apply **[COMBINATION FORM: Transformation via Subtraction/Recession]** to represent mitigation strategies, carbon sinks, or strict ecological budget compliance:

- **BLUE FLOOR:** Introduce an explicit structural breach, manifesting as a deep, subtractive **<Volume and Form>** hole in the exact center of the room. This negative space represents radical reduction pathways and structural carbon sinks.

3. Holistic Visual Synthesis & Cognitive Impact

- **Visual Balance:** Adjust the layout hierarchy so that high-impact architectural phases generate a highly visible, distorted visual "burden," while low-impact or sequestered carbon metrics show light, stable, and regular geometry.
- **Rendering Style:** Maintain a highly specialized hybrid style that reads as an **AI-interpreted architectural sketch**. Do not aim for clinical photorealism; prioritize a creative, tactile interpretation where abstract ecological boundaries are rendered as physical, felt attributes (mass, gradient, and density).

- **Final Output Composition:** Layer these distinct data-driven forms to ensure holistic superimposition—allowing non-expert stakeholders to perceive all quantitative material metrics and physical parts simultaneously in a single, glanceable view.

Final Image of Preliminary Test

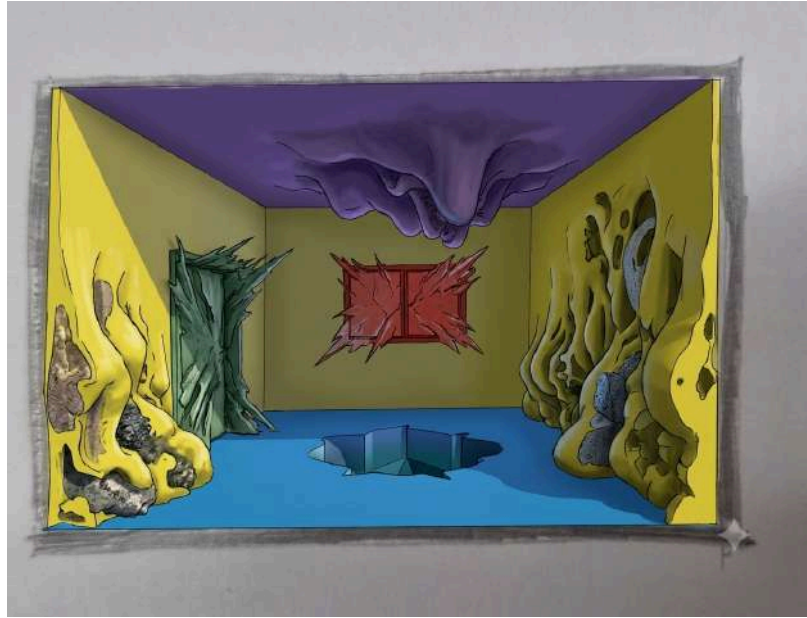


Figure 3. Final Image of Preliminary Exploratory Test. Source: Authors (2026)

2. The Baseline Layer and Prompting Strategy

Following the preliminary pilot test, the final strategy for prototyping was established, and the following workflow was applied to the prompt creation process.

A generic 3D interior model (Baseline Layer) was developed, in which the baseline components follow a specific color-coding: yellow slabs, pink walls, blue windows, and green doors.

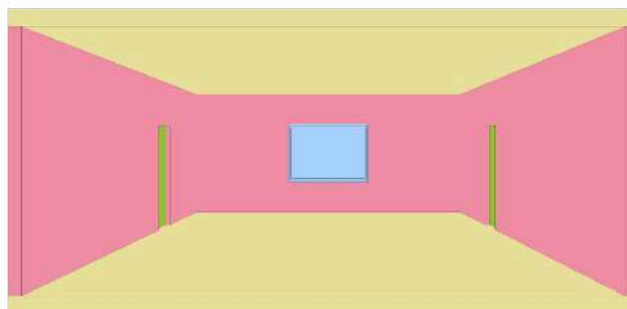


Figure 4. The Baseline Layer. Source: Authors (2026)

To represent EC through MET-VIS across different intensity scales for better comprehension of its behavior, an EC intensity hierarchy was established for each architectural component. The hierarchy, in descending order, is as follows: Walls > Top Slab > Bottom Slab > Window > Doors.

	Walls	Maximum EC Intensity
	Top Slab	High EC Intensity
	Bottom Slab	Moderate-low EC Intensity
	Window	Low EC Intensity
	Doors	Minimum EC Intensity

Based on the findings from the Systematic Literature Review in Phase 1 — specifically regarding the proposed metaphorical and visual principles — and the Graphic Structure developed, which culminated in the proposal of a 'Composite Metaphor,' the MET-VIS framework was developed based on the following conditions. Designated as 'Visual Encoding Rules,' these conditions served as the instructions for the image generation process.

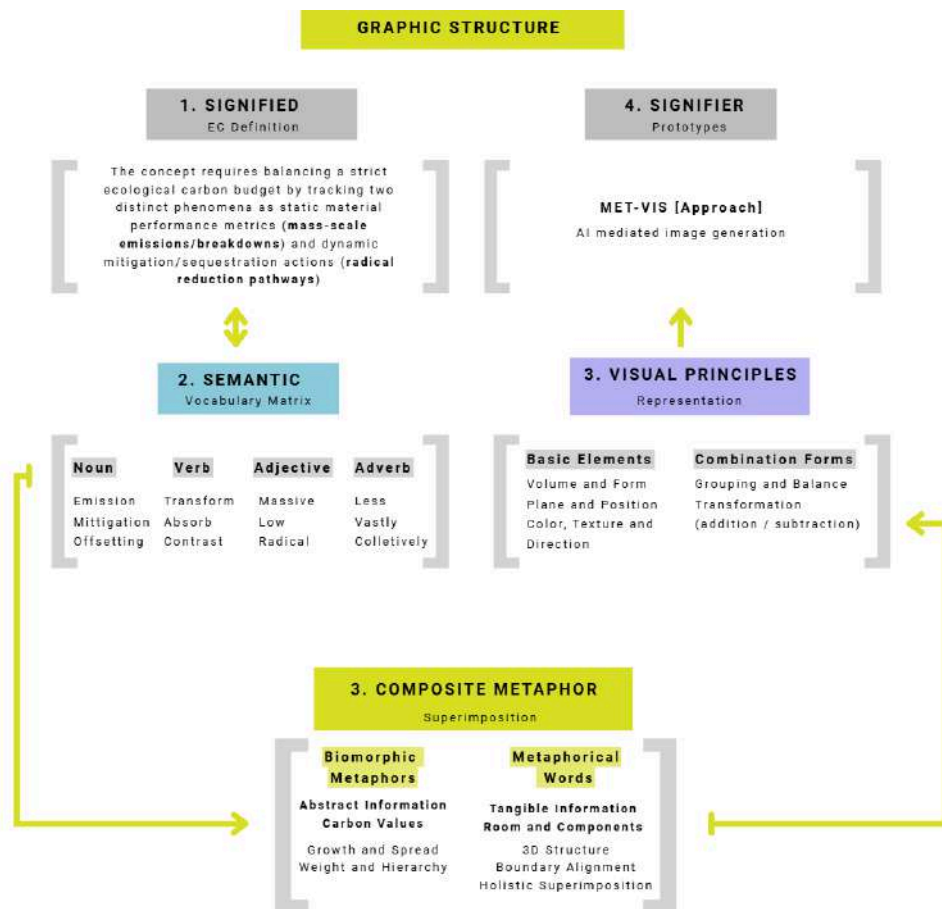


Figure 5. Graphic Structure definition. Source: Authors (2026)

Visual Encoding Rules

The Biomorphic Metaphor Layer	The prototype preserves the accurate architectural geometry of the component, while the "Metaphorical Layer" overlays data.
Opacity	Architectural volumes become semi-transparent; highlight only the edges and limits in their original colors to emphasize the presence of the biomorphic metaphor layer without abandoning the original geometry of the architectural structure.
Texture	An internal organic parasitic structure pattern resembling hexagonal rings or carbon chains is applied throughout the internal volume of each architectural component. Through this structure, we evoke the concept of carbon as a fluid, parasitic organism, achieving the intended "Composite Metaphor" (or Biomorphic Metaphor).
Color	Structures use a color gradient ranging from blue or green (low EC) to intense red (high EC). Colors are utilized as a supporting visual element to enhance the audience's cognitive fluency, as this strategy is academically reinforced and recognized for its intuitive ease of understanding.
Deformation	High EC causes structures to protrude from element boundaries as aggressive biomorphic spikes. This action evokes the use of natural behaviors—such as distortion—to represent abstract quantitative changes, which is effective as it taps into human sensory-motor responses to signal environmental impact, consistent with cases of "Naturalistic Metaphorical Mapping".

Prompting and baseline layer conceptualization

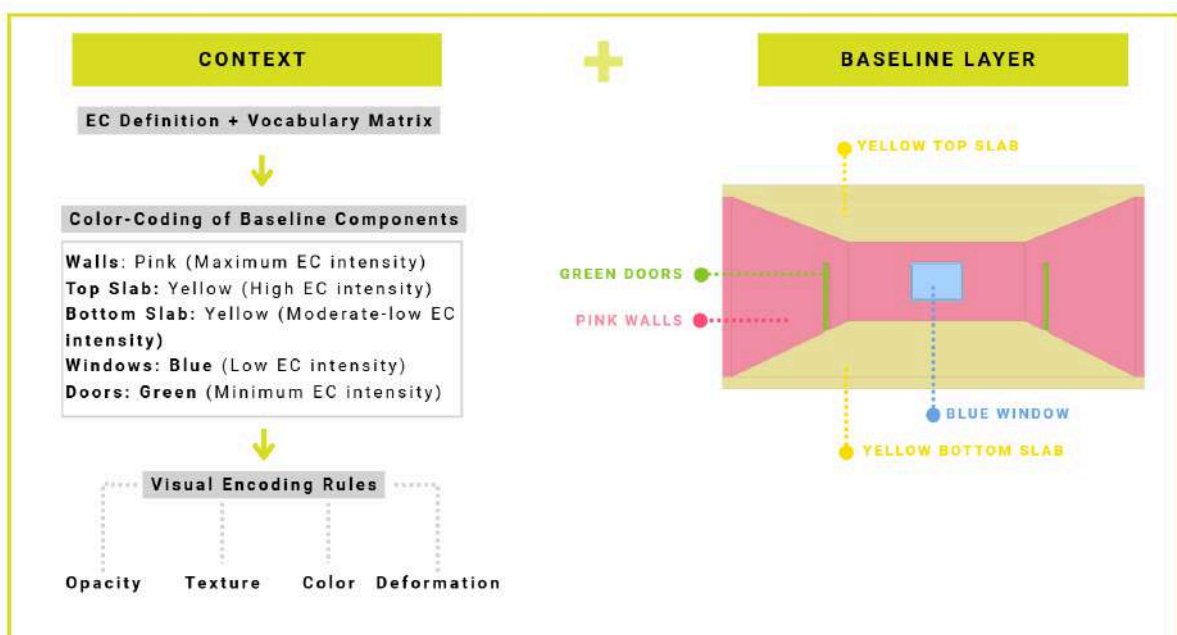


Figure 6. Prompting and baseline layer conceptualization. Source: Authors (2026)

Subsequently, the following prompting strategy was developed:

1°	Generic 3D interior modeling (Baseline Layer).
2°	Prompt drafting encoding MET-VIS requirements (EC context, and visual rules instructions using attention control and structural delimiters to highlight the vocabulary matrix and sequential steps).
3°	Prompt upload to Gemini for structural refinement.
4°	3D model image upload alongside the structured prompt to Firefly.
5°	Prompt improvement after identifying guideline gaps and inconsistencies.

Evaluation Criteria for Generated Prototypes: Generated outcomes were filtered through observational assessments focused on design syntax harmonization rather than a rigid checklist, verifying alignment between visual results, established rules, and the conceptual framework.

3. Restrictive Layer-based Test x Latent-based Test

Prior to conducting the formal prototyping tests, the need to categorize the evaluation into two distinct groups was identified: Restrictive Layer-based Tests and Latent-based Tests.

The initial set of tests, labeled "Restrictive Layer-based", utilized the full suite of rules, requiring iterations to align the graphic outcomes with the baseline hierarchy.

Conversely, "Latent-based" tests explore more open conditions: prompts retaining metaphorical definitions but with less controlled visual rules, producing a distinct MET-VIS concept.

4. Restrictive Layer-based Tests

First, the conceptualization of the prompt was developed:

Final Conceptualization

CONTEXT

<Embodied Carbon> is a <massive> environmental challenge characterized by <high-intensity>, carbon-intensive <impacts> across a <building's> lifecycle. The concept <requires balancing> a strict ecological carbon budget by <tracking two distinct phenomena> as STATIC MATERIAL PERFORMANCE METRICS (mass-scale emissions/breakdowns) and DYNAMIC MITIGATION/SEQUESTRATION ACTION (radical reduction pathways).

INSTRUCTIONS

Generate an <IMAGE> USING THE ATTACHED IMAGE AS BASELINE LAYER, for 3D metaphorical data visualization (MET-VIS) mapping abstract Embodied Carbon (EC) metrics onto a physical architectural container to bridge the affective gap and promote high cognitive fluency for non-experts.

The <attached image> shows a room that will be used to demonstrate the presence of different amounts of embodied carbon in the different architectural elements.

Take as a reference that in the <attached image>: yellow represents the top and bottom slabs, pink the walls, blue the windows, and green the doors.

For the carbon accounting representation example, in the image to be generated, we will assume that the embodied carbon intensity in descending order of intensity is: walls > top slab > bottom slab > window > doors. Consider the amount of embodied carbon in the walls as the highest possible degree of intensity and in the doors as the lowest degree. To represent this scenario, the following metaphorical representations will be used:

Superimpose a dynamic, data-driven <biomorphic metaphor> over the structural baseline to represent the <Massive environmental impact of Embodied Carbon>. The intensity of visual <biomorphic deformation> scales directly with numerical environmental severity (Higher EC Value = Higher Metaphorical Deformation).

The <biomorphic metaphor> layer:

Each architectural component (slabs, walls, windows, doors) from the baseline <attached image> will decrease in <opacity>, and what will be highlighted are the edges, corners, and limits of each element, which will remain in their original color.

The <volume of each element will house different quantities of organic, parasitic structures that resemble <hexagonal rings>, like carbon chains, metaphorically representing the presence of embodied carbon in the architectural components. The <colors> of these hexagonal parasitic structures *range from a light blue to an intense red gradient as the intensity of embodied carbon increases* in each element. Furthermore, as the embodied carbon intensity increases, the *hexagonal parasitic structures deform and protrude from the volume limits of each component* like aggressive biomorphic spikes.

Summarizing the visual elements and changes:

Opacity: the opacity of the architectural component decreases, and their volume becomes entirely perceptible from within; the edges and extremities of each component maintain their original color.

Texture: hexagonal parasitic structures inside the volumes of each element: greater quantity of hexagonal parasitic structures equals higher EC intensity.

Color: hexagonal parasitic structures will oscillate on a color gradient from light blue (lower EC intensity) to intense red (high EC intensity).

Deformation: hexagonal parasitic structures deform and protrude from the volume limits of each element like aggressive biomorphic spikes, according to the increase in EC intensity.

Remember: consider the amount of embodied carbon in the walls top slab as the highest possible degree of intensity and in the doors as the lowest degree.

Visual Synthesis & Cognitive Impact

Visual Balance: Adjust the layout hierarchy so that high-impact architectural phases generate a highly visible, distorted visual "burden," while low-impact or sequestered carbon metrics show light, stable, and regular geometry.

Rendering Style: Maintain a highly specialized hybrid style that reads as an AI-interpreted architectural sketch. <Do not aim for clinical photorealism>; prioritize a creative, tactile interpretation where abstract ecological boundaries are rendered as physical, felt attributes (mass, gradient, and density).

Final Output Composition: Layer these distinct data-driven forms to ensure holistic superimposition—allowing non-expert stakeholders to perceive all quantitative material metrics and physical parts simultaneously in a single, glanceable view.

AI-refined Prompt

Context & Core Subject:

Generate a 3D metaphorical data visualization (MET-VIS) of Embodied Carbon (EC) in a room interior. The image must represent a building lifecycle impact assessment, mapping carbon metrics onto a physical architectural container to demonstrate environmental performance.

Structural Baseline (Architectural Container):

Scene: An interior architectural space showing structural components.

Color-Coding of Baseline Components:

Top Slab: Yellow (Maximum EC intensity)

Walls: Pink (High EC intensity)

Bottom Slab: Yellow (Moderate-low EC intensity)

Windows: Blue (Low EC intensity)

Doors: Lilac (Minimum EC intensity)

Visual Transformation & Encoding Rules (Metaphorical Layer):

Opacity: All architectural volumes become semi-transparent, revealing their interior. Original colors are retained only for the edges and boundaries of each element.

Texture (Internal Growth): The interior of each volume is populated with parasitic hexagonal ring structures (carbon chain metaphor). The quantity of these structures corresponds directly to EC intensity.

Color Mapping: The hexagonal structures use a gradient color scale: Light Blue (low EC intensity) to Intense Red (high EC intensity).

Deformation: Geometric distortion scales with environmental severity. High EC intensity leads to the proliferation of aggressive biomorphic spikes protruding beyond the physical boundaries of the architectural element.

Intensity Hierarchy (Decending Order): Top Slab > Walls > Bottom Slab > Windows > Doors.

Holistic Visual Synthesis & Style:

Style: Hybrid AI-interpreted architectural sketch. Avoid clinical photorealism. Prioritize a creative, tactile interpretation where abstract ecological data is rendered as physical attributes (mass, gradient, and density).

Composition: Maintain a "glanceable" hierarchy where high-impact elements (Top Slab/Walls) create a distorted, aggressive visual burden, while low-impact elements (Doors) retain stable, regular geometry.

Goal: The final image must represent all quantitative metrics and physical parts simultaneously in a single, holistic 3D view.

Due to the character constraints imposed by Adobe Firefly, the prompts for all subsequent tests were synthesized based on the AI-refined prompt presented above.

4.1 Excluded Tests

The category 'Excluded Tests' refers to preliminary trials within the Restrictive Layer-based group that were omitted from this article, as they were subject to ongoing methodological refinement.

Excluded Test 1

Prompt:

Create an Image of 3D metaphorical visualization of Embodied Carbon (EC) in an architectural room. Baseline layer: Yellow slabs, Pink walls, Blue windows, Green doors. Intensity hierarchy: Walls > Top Slab > Bottom Slab > Window > Doors.

Visual Encoding Rules:

Opacity: Architectural volumes become semi-transparent; highlight only edges/limits in original colors.

Texture: Internal organic parasitic structures pattern (hexagonal rings/carbon chains) scale in quantity with EC intensity.

Color: Structures use a gradient from Blue or Green (low EC) to Intense Red (high EC).

Deformation: High EC causes structures to protrude from element boundaries as aggressive biomorphic spikes with Intense Red edges.

Image Result:

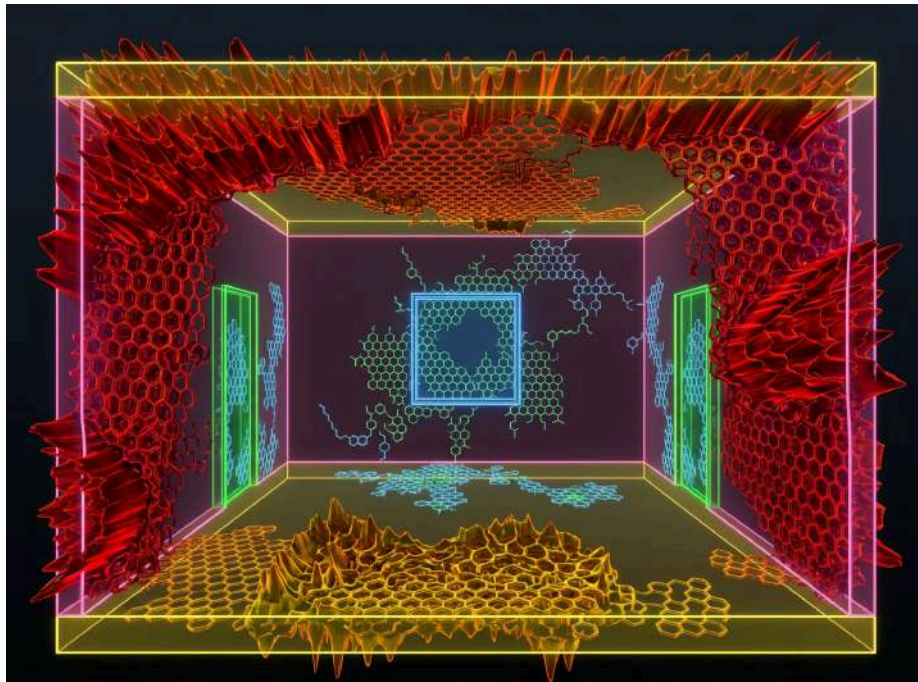


Figure 7. Image Result of Excluded Test 1. Source: Authors (2026)

Excluded Test 2

Prompt:

3D metaphorical visualization of Embodied Carbon (EC) in an architectural room. Baseline layer: Yellow slabs, Pink walls, Blue windows, Green doors. Intensity hierarchy: Top Slab > Walls > Bottom Slab > Window > Doors.

Visual Encoding Rules:

Opacity: Architectural volumes become semi-transparent; highlight only edges/limits in original colors.

Texture: Internal organic parasitic structures (hexagonal rings/carbon chains) scale in quantity with EC intensity and they are never non-existent.

Color: Structures use a gradient from Light Blue (low EC) to Intense Red (high EC).

Deformation: High EC causes structures to protrude from element boundaries as aggressive biomorphic spikes.

Addition: Remember: consider the amount of embodied carbon in the top slab as the highest possible degree of intensity and in the windows and doors as the lowest degree.

Image Result:

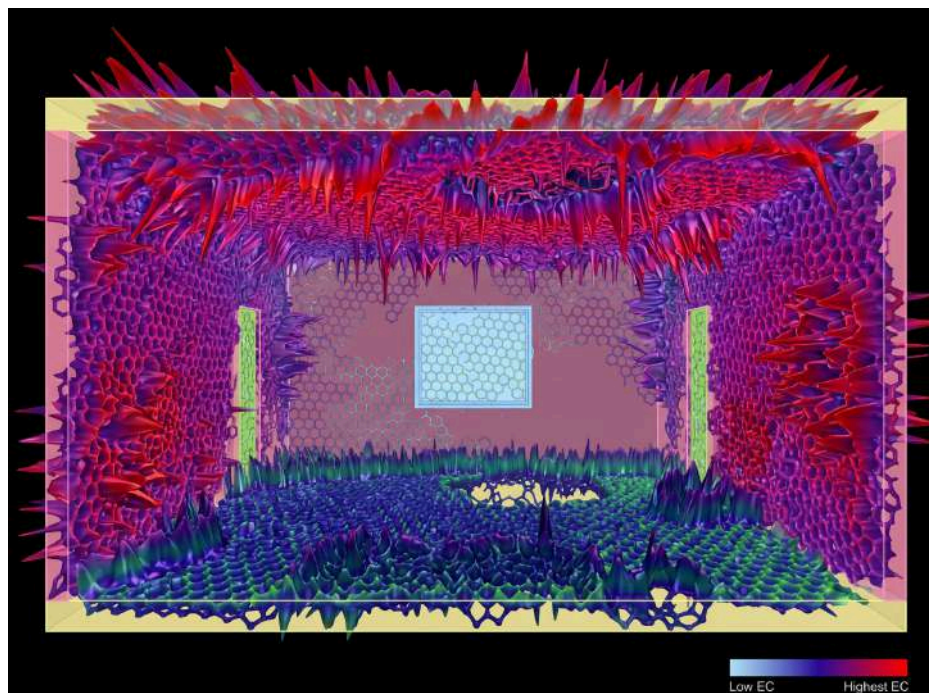


Figure 8. Image Result of Excluded Test 2. Source: Authors (2026)

4.2 Reference image

Due to the lack of alignment between the visualizations generated during the Excluded Tests and the proposed Visual Encoding Rules, it became necessary to develop a reference image to demonstrate the tangible representation of Embodied Carbon (EC) within architectural components.

The reference image was created without using prior visual references, based solely on the text prompt provided by the authors, which was derived from the EC definition, the vocabulary matrix, and the composite metaphor.

Prompt:

Produce an image of a rectangular surface viewed from above, as under a microscope, formed by a hexagonal mesh across its entire extent. This mesh resembles hexagonal rings like those of aromatic carbon chains, similar to the packing of spheres. The rings are small. This mesh follows the colors of the heat map (green and blue > red) in the hexagonal rings. Therefore, these rings have their edges and their interiors filled with shades of green and blue. In some irregular and random sections of the rectangular surface, each ring, in its inner region, deforms perpendicularly to the surface, becoming protuberances that resemble clusters of spikes. These spikes undergo an intense color gradient, with the hexagonal base varying from blue to intense red at the tips of the spikes. All spikes have red tips and bases close to the hexagonal rings in shades of green and blue; there are no blue spikes, and the tips of the spikes are always red, and the spikes are aggressive.

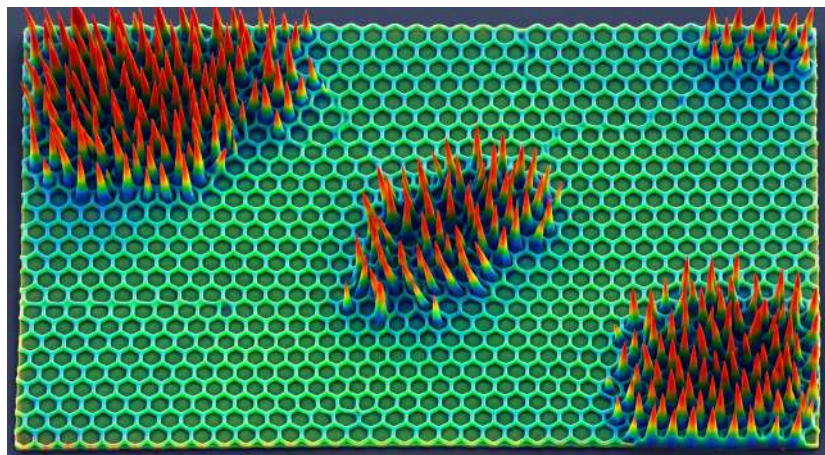


Figure 9. Reference Image. Source: Authors (2026)

Consequently, the tests conducted within the Restrictive Layer-based group incorporated the upload of this reference image alongside the Baseline Layer into Adobe Firefly.

Final Test

Create an Image of 3D metaphorical visualization of Embodied Carbon (EC) in an architectural room. Baseline layer: Yellow slabs, Pink walls, Blue windows, Green doors. Intensity hierarchy: Walls > Top Slab > Bottom Slab > Window > Doors. Visual Encoding Rules (follow the style of the second reference image to reproduce the visual encoding rules):

- Opacity: Architectural volumes become semi-transparent; highlight only edges/limits in original colors.
- Texture: Internal organic parasitic structures pattern (hexagonal rings/carbon chains).
- Color: Structures use a gradient from Blue or Green (low EC) to Intense Red (high EC).
- Deformation: High EC causes structures to protrude from element boundaries as aggressive biomorphic spikes with Intense Red edges.

Image Result:

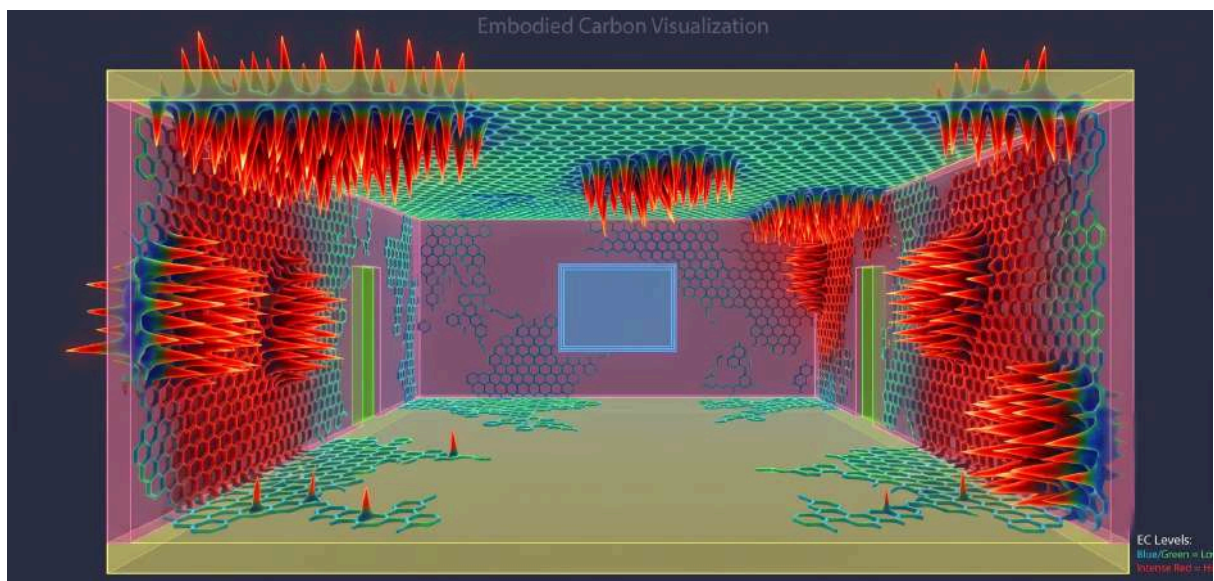


Figure 10. Initial Restrictive Layer-based Image. Source: Authors (2026)

The initial image from the Restrictive Layer-based test failed to adhere to certain pre-established instructions and required refinements, specifically: i) fully filling the architectural components with the texture pattern; and ii) aligning the color gradient and deformation with the visual hierarchy defined by the baseline layer, thus achieving the result presented in the Final Image.

Initial Test

Prompt:

Create an Image of 3D metaphorical visualization of Embodied Carbon (EC) in an architectural room. Baseline layer: Yellow slabs, Pink walls, Blue windows, Green doors. Intensity hierarchy: Walls > Top Slab > Bottom Slab > Window > Doors. Visual Encoding Rules (follow the style of the second reference image to reproduce these visual encoding rules):
Opacity: Architectural volumes become semi-transparent; highlight only edges/limits in original colors.
Texture: Internal organic parasitic structures pattern (hexagonal rings/carbon chains) along all the internal volume of each architectural component.
Color: Structures use a gradient from Blue or Green (low EC) to Intense Red (high EC).
Deformation: High EC causes structures to protrude from element boundaries as aggressive biomorphic spikes with Intense Red edges.

Image Result:

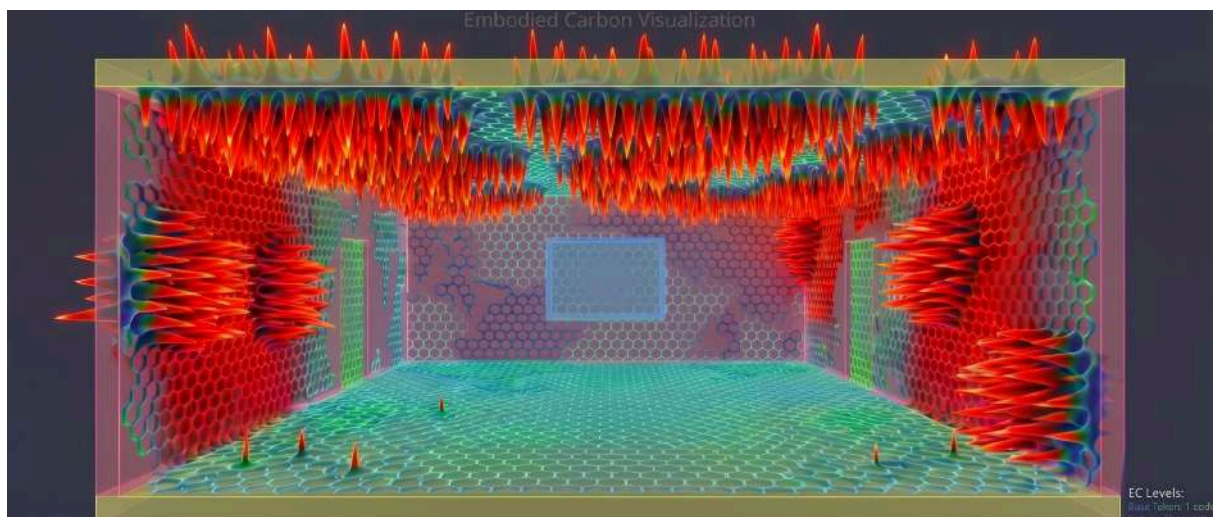


Figure 11. Final Restrictive Layer-based Image. Source: Authors (2026)

5. Latent-based Tests

For the Latent-based testing phase, an Initial Test and a Final Test were developed. The Initial Test established the following:

- Image generated based on the AI's interpretation of the provided context.
- No explanation of the Visual Encoding Rules was provided.
- Neither the Color-coded Components nor the EC intensity hierarchy was specified.
- The EC intensity hierarchy was specified.

Initial Test

Prompt:

3D metaphorical visualization of Embodied Carbon (EC) in an architectural room. Baseline layer: Yellow slabs, Pink walls, Blue windows, Green doors.

Visual Encoding Rules: Opacity, Texture, Color and Deformation scale in quantity with EC intensity.

Image Result:



Figure 12. Initial Latent-based Image. Source: Authors (2026)

Conversely, the Final Test determined:

- No explanation of the Visual Encoding Rules was provided.
- No Color-coded Components were introduced.

Final Test

Prompt:

3D metaphorical visualization of Embodied Carbon (EC) in an architectural room. Baseline layer: Yellow slabs, Pink walls, Blue windows, Green doors. Intensity hierarchy: Walls > Top Slab > Bottom Slab > Window > Doors.

Visual Encoding Rules:

Opacity, Texture, Color, Deformation scale in quantity with EC intensity.

Image Result:

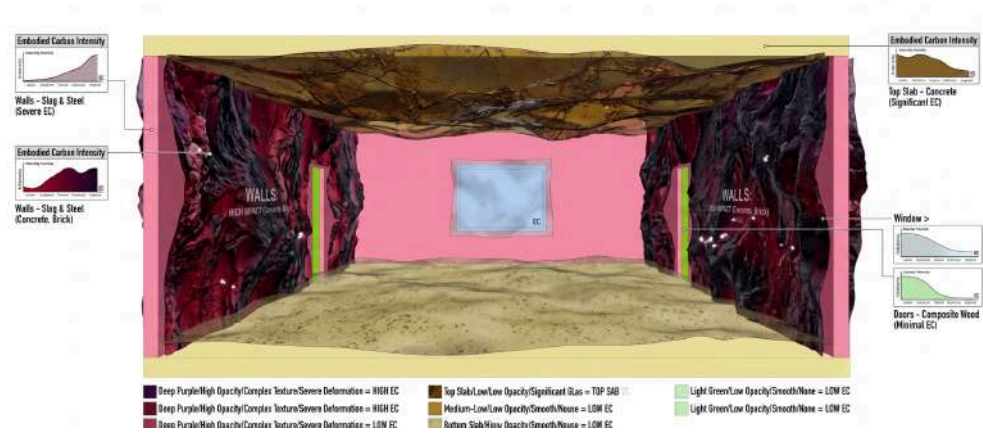


Figure 12. Final Latent-based Image. Source: Authors (2026)